Final Project Report
Padmalekha (pdlekhaa)
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Introduction: Criminal activity is difficult to predict, but is a necessary tool to maintain a safe society. Having a greater understanding allows for targeted approaches to reduce crime, such as government programs or education/rehabilitation funding.

I am trying to solve the problem of estimating/narrowing the underlying parameter which is the proportion of crimes that are done by women compared to men, and want to employ techniques to help improve this estimate. Previous work regarding crime and Bayesian data analysis has focused on spatial data, such as Hu et al. [2018] who performed cluster work in space and time and Law and Haining [2010] who modeled crime areas in Sheffield, rather than demographic information.

Dataset: I'm using a dataset from Kaggle called "San Jose Police Arrests" which details each of the arrests, charges, and incidents in San Jose from 2015-2022 Sherman [2022]. I focused on the arrests dataset, in which each row has information about the $AGE, SEX, RACE_CODE, DATE$ of an arrest. This was joined with the incidents dataset where available, describing a corresponding "incident" which caused the arrest. Each incident has information $TIME, INCIDENT_DESCRIPTION$ that lead to the arrest.

The dataset had some non male/female values, counts are shown in Table 1, but because there were so few (<1%), these rows were dropped. I also joined another 2 datasets from the same Kaggle resource to map $RACE_CODE \rightarrow RACE \rightarrow RACE_GROUP$, counts shown in Table 2. Additionally, I created the column $DECADE$, which places a person's age into a bucket of size 10 years. For example, someone of age 55 would have $DECADE$ 50. The final dataset (2015-2020) had 134,683 rows, with examples shown in Figure 1. Note: 2018 was missing entirely, and 2021 data was held out for prediction.

Methods: My analysis contained three main methods to identify the best posterior that could have generated the data. Firstly, I explored a beta-binomial one-parameter Bayesian analysis. Next, I investigated hierarchical structures in the dataset, and finally, I tested a sequential Bayes method.

Model Assumption
To begin, I checked the distribution of age when looking at a histogram with only female arrests and only male arrests. The histograms follow roughly the same distribution for AGE, with different magnitudes Figure 2. There was also not a huge difference in SEX vs $RACE_GROUP$. So, I am assume exchangeability, meaning that reordering the outcome would not change the results.

ONE PARAMETER MODELS

Conjugate Prior
I explored a closed form conjugate distribution for the data as a baseline. I chose a binomial distribution for the sampling model/data. This means the dataset represents a count (total female) out of *N* trials (total arrests). For the prior probability, I chose the conjugate beta prior. The National Bureau of Justice Statistics has reported that female incarceration rates are usually around 22% U.S. Department of Justice, Bureau of Justice Statistics [1999], informing my prior to be Beta$(2, 7)$.

Hyperparameter search on prior with MCMC
I also experimented with changing the prior hyperparameters within the beta distribution shown in Table 3, where the expected value remained around 0.2-0.22, using the MCMC method in Stan.

Checks
The conjugate posterior has a mean of 0.2353, which is very similar to the sampling model, because the number of rows is so high. I performed posterior predictive checks using the groups of decade (and race group), but this posterior does not do a good job of prediction, because the true mean of each group is not always in the realm of possibilities found by the predictive distribution. In fact, it only fits well meaningfully with people in their 30s and in their 80s, shown in Figure 3. This implies that the initial exchangability assumption might not actually make sense.

As for the MCMC hyperparameter search, the HDIs and 90% credible intervals in Figure 4 show that the search does not yield very different results, so I continue to use Beta$(2, 7)$.

HIERARCHICAL MODELS

The checks from the previous one-parameter model informed my decision to incorporate hierarchy from the structures in the dataset. I fit three hierarchical models to investigate if they improved the model. I began with using $RACE_GROUP$ (6 blocks) and $DECADE$ (10 blocks) by adding parameter κ which represents the overall concentration, and is specified by $\kappa \sim$ Gamma$(0.01, 0.01)$. I kept $\mu \sim$ Beta$(2, 7)$, which is the overall shared mean. I lastly fit a hierarchical model using time, turning the $TIME$ of an incident into $TIME24HR$, which take a value from 0-23. These represent 1 hour time blocks, where 3:45pm becomes 15, for example.

Checks
All of the estimates converged with Rhat values of 1. The hierarchical model with the most variability was the hierarchy on decade, shown in Figure 5. This may be due to the large variability in the number of people in each decade block (there are very few arrests of people between the ages of 0-10, 80-90, and 90-100).

Surprisingly, the posterior predictive checks using the hierarchical models on race group and decade did not improve when incorporating these into the hierarchy. For example, Figure 6 and Figure 7 show a poor posterior predictive check with the histogram not accommodating the true proportion.

The time blocks hierarchical model fit the data much better when doing a posterior predictive check on both $DECADE$, Figure 8 and on $RACE_GROUP$ Figure 9. I checked a few different hours and the estimate was almost always in the distribution, even with only the 100 samples. However some hours, like hour 1, in the hierarchical model did not perform very well. It seems like nighttime hours are worse in the posterior predictive check by decade. However, the time hierarchical model κ value was very large, around 558, meaning that there is little variability with the θ estimates.

SEQUENTIAL MODEL

Lastly, I used the best performing hierarchical model to fit a sequential model. This meant choosing the original prior and performing the hierarchical beta-binomial MCMC on only 2015 data, but then using the 24 θ estimates (one per hour) as part of the prior for the next year (mus), and so on. Each next θ (24 values) followed

$$\theta \sim \text{beta_proportion(mus, } \kappa)$$

where the mus are the previous years mean estimate of each θ block. There was no shared mean after the first fit. The final fit was on the 2020 data.

Checks
This model performed well on various hours when doing a PPC check on both decade Figure 10and race group Figure 11. The κ value is much lower than the non-sequential hierarchical model at 77.91.

Results: After running each of these models and iteratively improving them based on predictive posterior checks, I finally compared all of them using 3 metrics: runtime, LOO estimates, and predictive performance.

Unsurprisingly, the runtime table Figure 12 shows that the hyperparameter search takes the longest amount of time by far, and this did not result is any improvement of the parameter. The next longest was the sequential hierarchical model, while the rest were about the same.

I also found scores for ELPD, P_LOO, and LOOIC using the `loo` package for Bayesian models in Figure 13. They actually seem to agree that the MCMC simple models are the best, because they have a high ELPD (estimated predictive accuracy) and a low LOOIC (similar to AIC), along with of course small estimated parameters (P_LOO).

Finally, actual prediction was done on the heldout 2021 dataset. For this, each model fit outputted a distribution (standalone, or based on time, decade, or race group) and the true proportion was displayed on top of the histogram. For the single-parameter models, the prediction was very far off (Figure 14) as well as the hierarchical models on $DECADE$, and $RACE_GROUP$ (Figure 15).

The best model for the prediction on 2021 was by far the sequential model (which uses the time block hierarchy). For each hour, it had an accurate distribution of θ which included the true θ from 2021, Figure 16 and Figure 17.

Conclusion: It is clear that there are techniques to estimate the proportion of female arrests in a given year, particularly by following a sequential pattern year after year, and using a hierarchy by hour. Despite the improvements in PPCs and actual prediction, I learned that not all metrics agreed with this (LOO estimates). Narrowing down these estimates is extremely helpful in deciding where to spend money and other methods to reduce crime, which may correlate with working hours, daylight hours, etc. even by sex.

Some shortcomings of my project was a lack of investigation of different prior distributions for θ as well as κ which may have yielded more interesting results. I also could have explored interactions with variables more, which is challenging to incorporate in Bayesian data analysis, especially while retaining a hierarchical structure. In the future with more time or resources, I would improve upon these limitations and potentially add an element of location/district in San Jose, which may be even more informative.

Figures and Tables

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36990	125165	43	118

Table 1: Counts of SEX in Arrests dataset

RACE GROUP	Count
AFRICAN AMERICAN	18927
ASIAN/PACIFIC ISLANDER	16113
CAUCASIAN	29100
HISPANIC/LATIN/MEXICAN	85626
OTHER	2456
UNKNOWN	816

Table 2: Counts of $RACE_GROUP$ in Arrests dataset

	PIN <int>	INCIDENT.NO <chr>	AGE <dbl>	SEX <chr>	RACE_GROUP <chr>	RPT.TIME <int>	INCIDENT.DESCRIPTION <chr>	YEAR <int>	DECADE <dbl>
89080	442029349	163520618	14	M	HISPANIC/LATIN/MEXICAN	1830	11357A HS POSN CONC CANNABIS	2016	10
27605	442009014	162320138	24	M	CAUCASIAN	420	12020(A)PC POSN DANGEROUS WEAP	2016	20
33788	441896879	151680053	32	M	CAUCASIAN	332	23152B VC DUI ALCOHOL .08%	2015	30

Figure 1: Sample of 3 rows from dataset

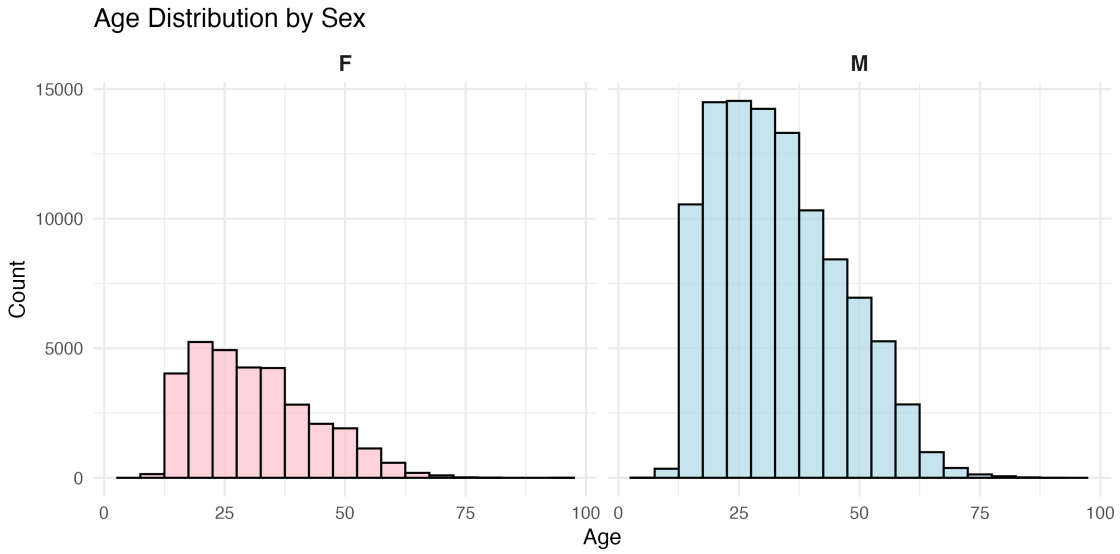


Figure 2: Histograms of AGE by SEX

α	β
2	7
10	45
22	78
154	700

Table 3: Values tested as hyperparameters on Beta(α, β) prior

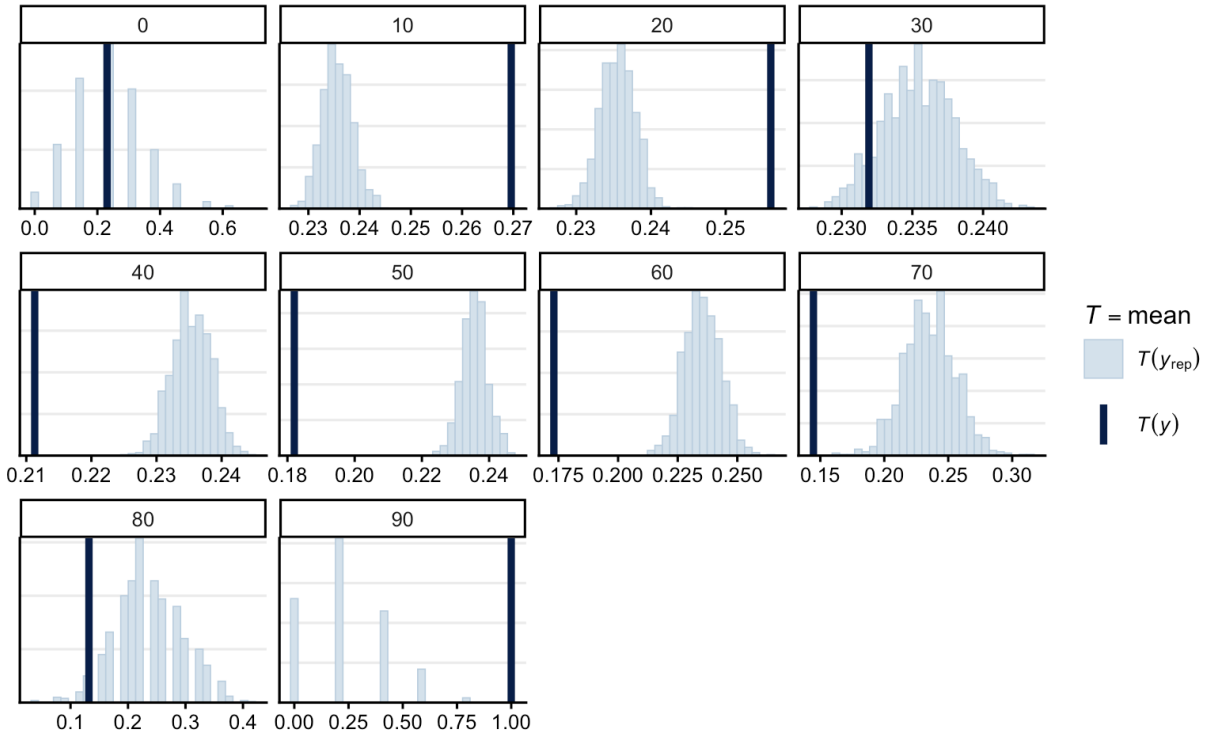


Figure 3: Posterior Predictive Check on Closed Form Model

alpha <dbl>	beta <dbl>	mean_theta <dbl>	sd_theta <dbl>	hdi_lower <dbl>	hdi_upper <dbl>
2	7	0.235	0.001	0.233	0.238
10	45	0.235	0.001	0.233	0.238
22	78	0.235	0.001	0.233	0.238
154	700	0.235	0.001	0.233	0.237

Figure 4: Hyperparameter Search Results

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
theta[1]	0.22	0	0.06	0.12	0.18	0.22	0.26	0.35	3859	1
theta[2]	0.27	0	0.00	0.26	0.27	0.27	0.27	0.28	4741	1
theta[3]	0.26	0	0.00	0.25	0.25	0.26	0.26	0.26	5476	1
theta[4]	0.23	0	0.00	0.23	0.23	0.23	0.23	0.24	5313	1
theta[5]	0.21	0	0.00	0.21	0.21	0.21	0.21	0.22	4285	1
theta[6]	0.18	0	0.00	0.18	0.18	0.18	0.18	0.19	4093	1
theta[7]	0.17	0	0.01	0.16	0.17	0.17	0.18	0.19	4674	1
theta[8]	0.15	0	0.02	0.12	0.14	0.15	0.16	0.19	3326	1
theta[9]	0.17	0	0.04	0.10	0.14	0.17	0.20	0.26	4349	1
theta[10]	0.32	0	0.11	0.18	0.24	0.29	0.36	0.60	1445	1

Figure 5: θ Results from Hierarchical Model on $DECADE$

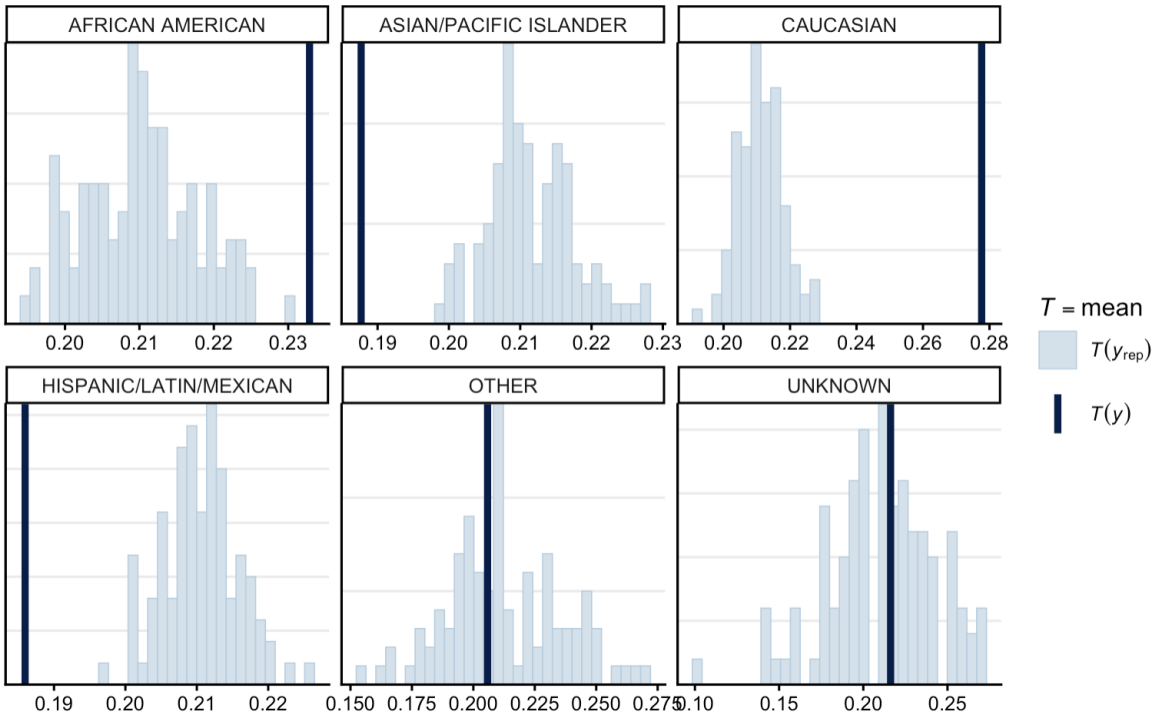


Figure 6: Hierarchical Model on $DECADE$, performing a PPC on $RACE_GROUP$, using θ where $DECADE = 5$

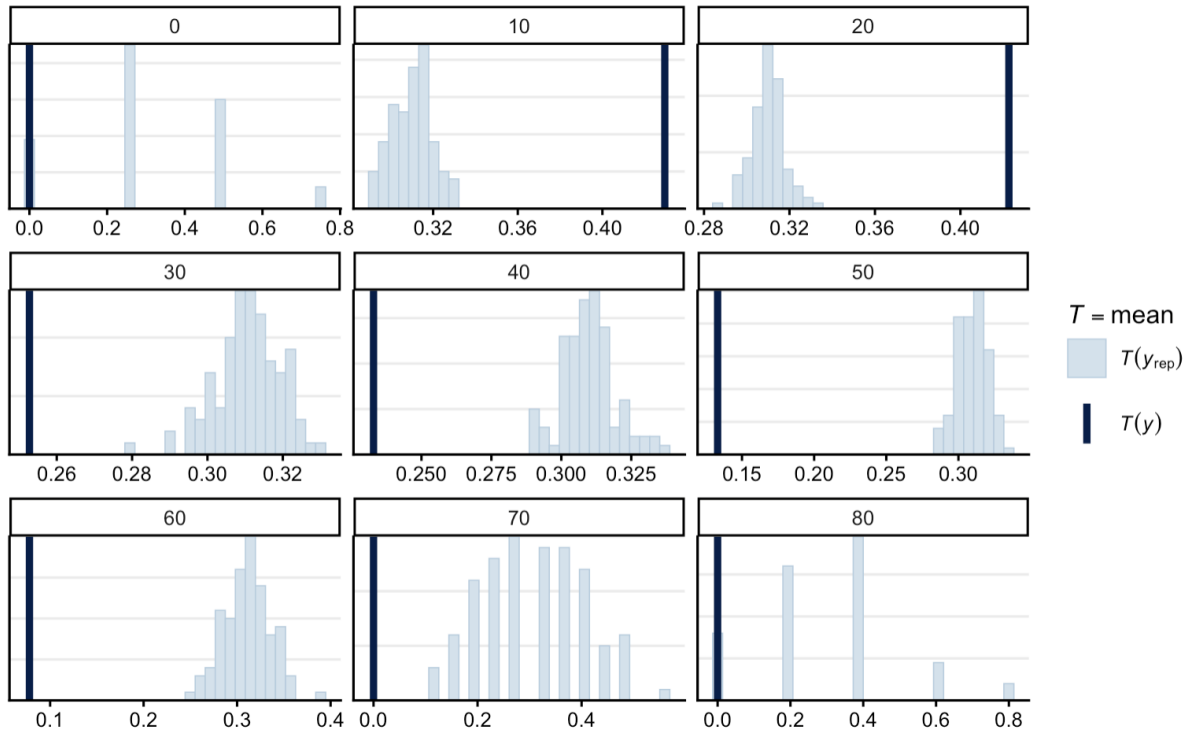


Figure 7: Hierarchical Model on $RACE_GROUP$, performing a PPC on $DECADE$, using θ where $RACE_GROUP$ = "African American"

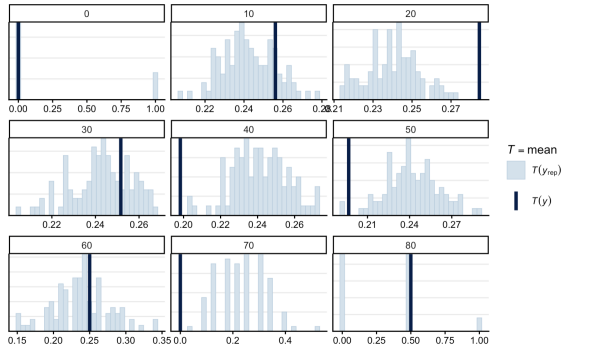


Figure 8: PPC for Hierarchical Model on Time Blocks, for group $DECADE$, using $TIME24HR$ = 14 (1-2pm)

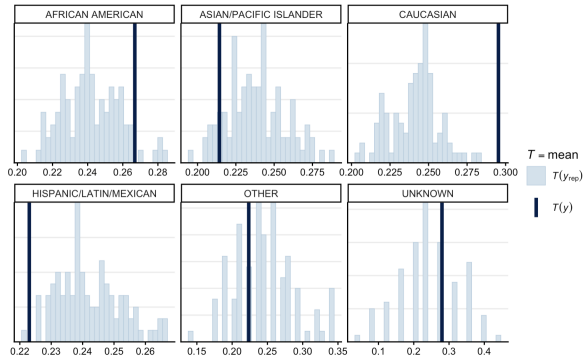


Figure 9: PPC for Hierarchical Model on Time Blocks, for group $RACE_GROUP$, using $TIME24HR$ = 14 (1-2pm)

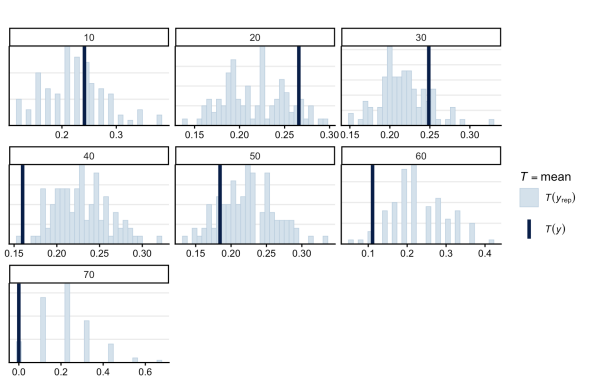


Figure 10: PPC for Sequential + Hierarchical Model (2020) on Time Blocks for group $DECADE$, using $TIME24HR$ = 14 (1-2pm)

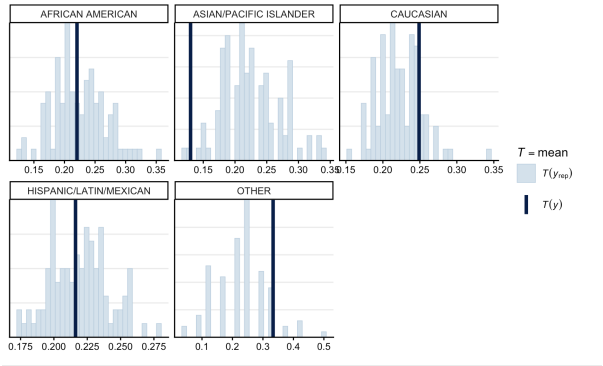


Figure 11: PPC for Sequential + Hierarchical Model (2020) on Time Blocks for group $RACE_GROUP$, using $TIME24HR$ = 14 (1-2pm)

	user.self	sys.self	elapsed	user.child	sys.child
closed_theta_time	0.019	0.003	0.042	0.000	0.000
mcmc1fit_time	0.165	0.014	0.274	0.000	0.000
mcmc_hyperparam_time	1.455	0.487	92.559	76.846	5.291
hier_hour_time	0.562	0.025	0.591	0.000	0.000
hier_race_time	0.323	0.008	0.330	0.000	0.000
hier_age_time	0.509	0.006	0.516	0.000	0.000
sequential_time	2.675	0.138	2.829	0.000	0.000

Figure 12: Runtime of Evaluating Each Model

Model <chr>	ELPD_LOO <dbl>	P_LOO <dbl>	LOOIC <dbl>
MCMC (2, 7)	-7.592351	1.2867900	15.18470
MCMC (10, 45)	-7.031302	0.7108749	14.06260
MCMC (22, 78)	-7.462150	1.1312410	14.92430
MCMC (154, 700)	-7.690937	1.3494467	15.38187
Hierarchical by Hour	-129.598655	17.6791857	259.19731
Hierarchical by Race	-33.914057	3.8765872	67.82811
Hierarchical by Age (10s)	-51.333790	8.2569116	102.66758
Sequential + Hierarchical by Hour	-110.699085	18.9019123	221.39817

Figure 13: Model Prediction with LOO estimates

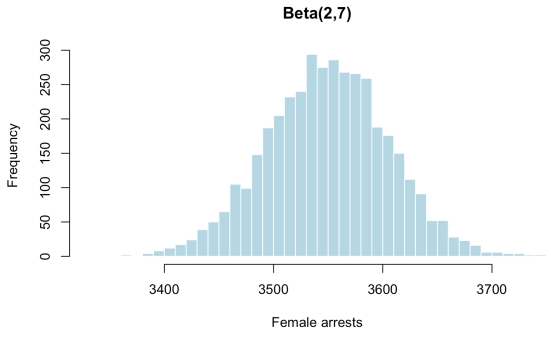


Figure 14: Prediction on 2021 dataset using One-Parameter Model where the true y is 2957 (not pictured)

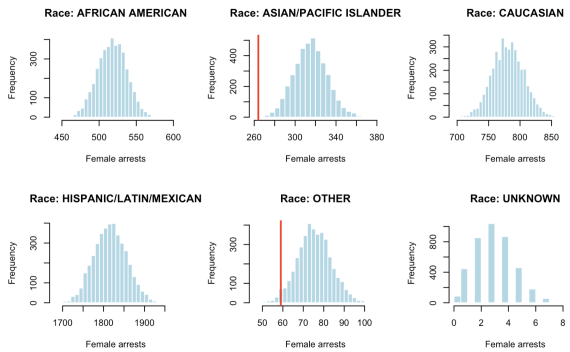


Figure 15: Prediction on 2021 Dataset using Hierarchical model on $RACE_GROUP$

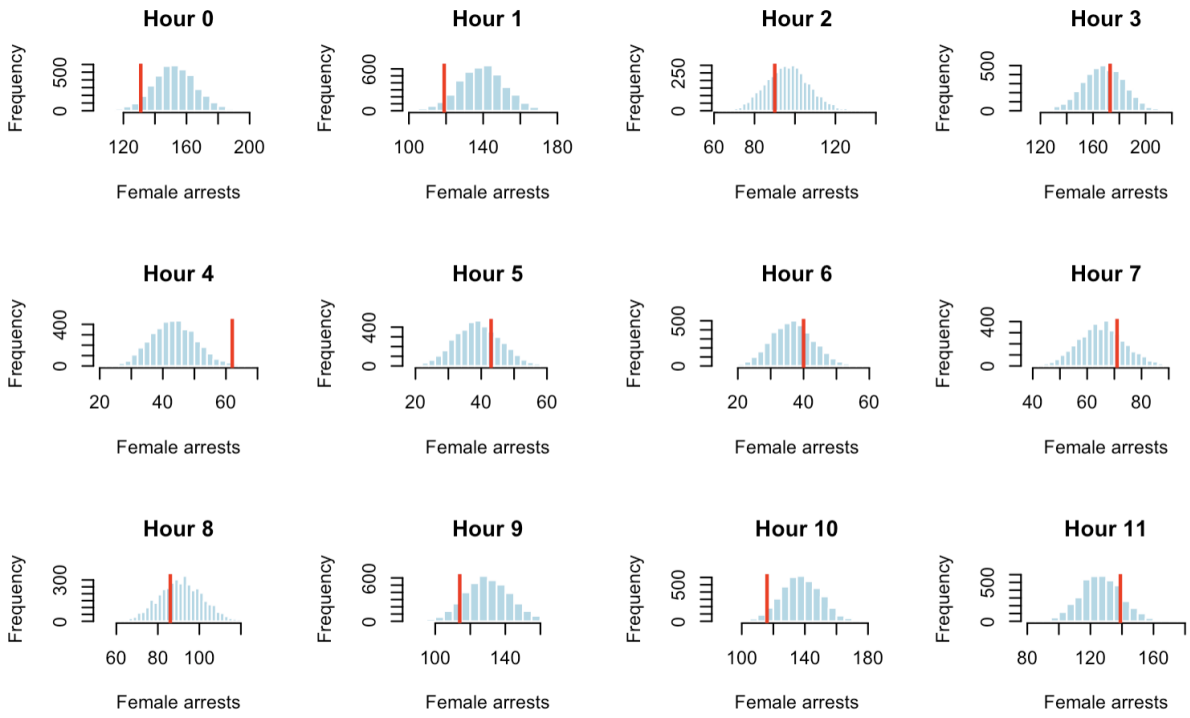


Figure 16: Prediction on 2021 dataset using Sequential Hierarchical Model on $TIME24HR$ (Hours 0-11, Midnight-Noon)

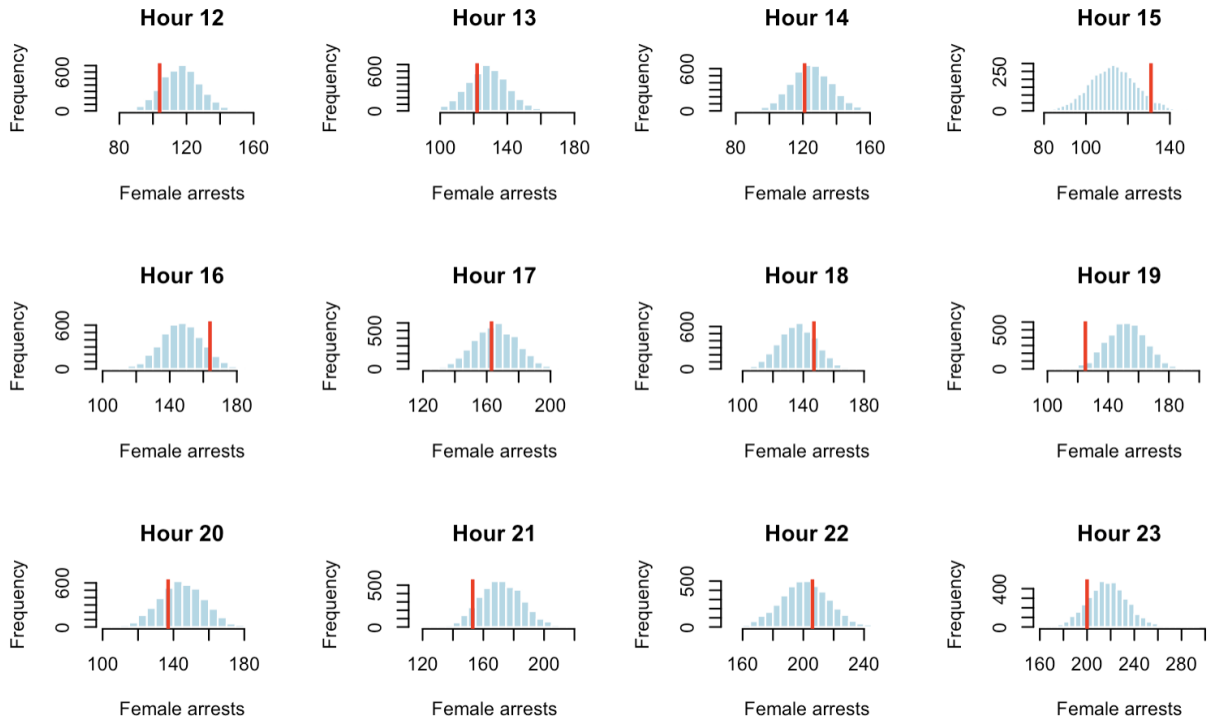


Figure 17: Prediction on 2021 dataset using Sequential Hierarchical Model on $TIME24HR$ (Hours 12-23 (Noon-Midnight)

References

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